

TARGET-90

JKBOSE CLASS-11TH

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BINOMIAL THEOREM

One Shot

MATHS

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Topics To Be Covered

- 1 ... Binomial Theorem (statement only)
- 2 ... Important Questions.
- 3 ...

BINOMIAL THEOREM



For any real numbers a and b and a positive integer n ,

$$(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$$

where ${}^nC_r = \frac{n!}{r!(n-r)!} \quad (0 \leq r \leq n)$

Expansion Examples

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

Important Points

- ★ The general term: $T_{r+1} = {}^nC_r a^{n-r} b^r$
- ★ Number of terms in the expansion: $n + 1$
- ★ Sum of coefficients: $(a + b)^n \rightarrow 2^n$
- ★ Middle term (for even n): $T_{(n/2)+1} = {}^nC_{n/2} a^{n/2} b^{n/2}$
- ★ Middle term (for odd n): Two middle terms: $T_{(n-1)/2+1}$ and $T_{(n+1)/2+1}$

Pascal's Triangle



Coefficients of $(a + b)^n$

$(a + b)^0$	$\rightarrow$	1
$(a + b)^1$	$\rightarrow$	1 1
$(a + b)^2$	$\rightarrow$	1 2 1
$(a + b)^3$	$\rightarrow$	1 3 3 1
$(a + b)^4$	$\rightarrow$	1 4 6 4 1
$(a + b)^5$	$\rightarrow$	1 5 10 10 5 1

INTRODUCTION



Binomial Theorem for Positive Integral Indices

Suppose that a and b be any two real numbers, then

$$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_n b^n$$

$$\Rightarrow (a+b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r \quad (\text{Binomial Theorem})$$

e.g: $(a+b)^2 = {}^2C_0 a^2 + {}^2C_1 ab + {}^2C_2 b^2$.

$$\Rightarrow (a+b)^2 = a^2 + 2ab + b^2.$$

$$* (a+b)^3 = {}^3C_0 a^3 + {}^3C_1 a^2 b + {}^3C_2 a b^2 + {}^3C_3 b^3.$$

$$\Rightarrow (a+b)^3 = a^3 + 3a^2 b + 3ab^2 + b^3.$$

* The number of terms in the expansion of $(a+b)^n$ is $(n+1)$.

* General term of the expansion of $(a+b)^n$ is

$$T_{r+1} = {}^nC_r a^{n-r} b^r.$$

EXAMPLE



Expand $\left(x^2 + \frac{3}{x}\right)^4$, $x \neq 0$

$$\frac{4!}{2!2!} = \frac{\overset{2}{\cancel{4}} \times 3 \times \cancel{2}}{\cancel{2}! \times \cancel{2}!}$$

(Sol) From Binomial Theorem, we have;

$$\left(x^2 + \frac{3}{x}\right)^4 = {}^4C_0(x^2)^4 + {}^4C_1(x^2)^3\left(\frac{3}{x}\right) + {}^4C_2(x^2)^2\left(\frac{3}{x}\right)^2 + {}^4C_3(x^2)\left(\frac{3}{x}\right)^3 + {}^4C_4\left(\frac{3}{x}\right)^4$$

$$\Rightarrow \left(x^2 + \frac{3}{x}\right)^4 = x^8 + 4x^6 \times \frac{3}{x} + 6x^4 \times \frac{9}{x^2} + 4x^2 \times \frac{27}{x^3} + \frac{81}{x^4}$$

$$\Rightarrow \left(x^2 + \frac{3}{x}\right)^4 = x^8 + 12x^5 + 54x^2 + \frac{108}{x} + \frac{81}{x^4}$$

EXAMPLE



Which is larger $(1.01)^{1000000}$ or 10,000?

H/W

EXAMPLE

$${}^nC_{n-1} = \frac{n!}{(n-1)!(n-(n-1))!} = \frac{n!}{(n-1)!1!} = n$$
$$9 = 4 \times 2 + 1$$
$$a = bq_r + r$$

Using binomial theorem, prove that $6^n - 5n$ always leaves remainder 1 when divided by 25.

Proof: From Binomial Theorem, we have;

$$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_{n-1} a b^{n-1} + {}^nC_n b^n \quad \text{--- (1)}$$

Setting $a=1$ and $b=5$ in (1), we get;

$$(1+5)^n = (1)^n + {}^nC_1 (1)^{n-1} (5) + {}^nC_2 (1)^{n-2} (5)^2 + \dots + {}^nC_{n-1} (1) (5)^{n-1} + (5)^n$$

$$\Rightarrow 6^n = 1 + 5n + 5^2 \left\{ {}^nC_2 + {}^nC_3 (5) + \dots + {}^nC_{n-1} (5)^{n-3} + (5)^{n-2} \right\}$$

$$\Rightarrow 6^n - 5n = 25q + 1$$

This shows that $6^n - 5n$ leaves the remainder 1 when divided by 25.

Hence proved.

QUESTION (NCERT EXAMPLAR)



The coefficient of x^p and x^q (p and q are positive integers) in the expansion of $(1+x)^{p+q}$, are

$$(a+b)^n \rightarrow T_{r+1} = {}^nC_r a^{n-r} b^r$$

1 Equal

$$(1+x)^{p+q} \rightarrow T_{r+1} = {}^{p+q}C_r x^r$$

2 Equal with opposite signs

3 Reciprocal of each other

4 None of these

Here Co-efficient of $x^p = {}^{p+q}C_p = \frac{(p+q)!}{p! \cdot (p+q-p)!} = \frac{(p+q)!}{p! \cdot q!}$

Co-efficient of $x^q = {}^{p+q}C_q = \frac{(p+q)!}{q! \cdot (p+q-q)!} = \frac{(p+q)!}{p! \cdot q!}$

QUESTION (NCERT EXAMPLAR)

The ratio of the coefficient of x^{15} to the term independent of x in $\left(x^2 + \frac{2}{x}\right)^{15}$ is

1 $\times$ 12 : 32

2 $\checkmark$ 1 : 32

3 $\times$ 32 : 12

4 $\times$ 32 : 1

(Sol) General Term of the expansion;

$$T_{r+1} = {}^{15}C_r (x^2)^{15-r} \left(\frac{2}{x}\right)^r$$

$$\Rightarrow T_{r+1} = {}^{15}C_r x^{30-2r} \cdot \frac{2^r}{x^r}$$

$$\Rightarrow T_{r+1} = {}^{15}C_r 2^r x^{30-3r}$$

For term independent of x , we have;

$$30 - 3r = 0$$

$$\Rightarrow r = 10$$

So Term Independent of $x = T = {}^{15}C_{10} 2^{10}$

For the Co-efficient of x^{15} , we have

$$30 - 3r = 15$$

$$\Rightarrow 3r = 15$$

$$\Rightarrow r = 5$$

$$\text{Co-efficient of } x^{15} = {}^{15}C_5 2^5$$

$${}^{15}C_5 = {}^{15}C_{15-5} = {}^{15}C_{10}$$

$$\text{Required ratio} = \frac{{}^{15}C_5 2^5}{{}^{15}C_{10} 2^{10}}$$

$$= \frac{1}{2^5} = 1 : 32$$

EXAMPLE



Determine whether the expansion of $\left(x^2 - \frac{2}{x}\right)^{18}$ will contain a term containing x^{10} ?

(Sol) General Term of the expansion of $\left(x^2 - \frac{2}{x}\right)^{18}$ is: For term containing x^{10} , we have;

$$T_{r+1} = {}^{18}C_r (x^2)^{18-r} \left(-\frac{2}{x}\right)^r$$

$$\Rightarrow T_{r+1} = {}^{18}C_r x^{36-2r} \frac{(-2)^r}{x^r}$$

$$\Rightarrow T_{r+1} = {}^{18}C_r (-2)^r x^{36-3r}$$

$$36 - 3r = 10$$

$$\Rightarrow 3r = 26$$

$$\Rightarrow r = \frac{26}{3} \notin \mathbb{N}, \text{ which is not possible.}$$

This shows that $\left(x^2 - \frac{2}{x}\right)^{18}$ do not contain a term with x^{10} .

Hence the result.

EXAMPLE



Using Binomial Theorem, indicate which number is larger $(1.1)^{10000}$ or 1000.

h/w

EXAMPLE

$$(a-b)^4 = (a+(-b))^4$$



Find $(a+b)^4 - (a-b)^4$. Hence, evaluate $(\sqrt{3} + \sqrt{2})^4 - (\sqrt{3} - \sqrt{2})^4$.

(Sol) From Binomial Theorem, we have;

$$(a+b)^4 = {}^4C_0 a^4 + {}^4C_1 a^3 b + {}^4C_2 a^2 b^2 + {}^4C_3 a b^3 + {}^4C_4 b^4.$$

$$\Rightarrow (a+b)^4 = a^4 + 4a^3 b + 6a^2 b^2 + 4ab^3 + b^4 \quad \text{--- (1)}$$

$$\text{Also } (a-b)^4 = (a+(-b))^4 = a^4 + 4a^3(-b) + 6a^2(-b)^2 + 4a(-b)^3 + (-b)^4$$

$$\Rightarrow (a-b)^4 = a^4 - 4a^3 b + 6a^2 b^2 - 4ab^3 + b^4 \quad \text{--- (2)}$$

Subtracting (1) & (2), we get;

$$(a+b)^4 - (a-b)^4 = 8a^3 b + 8ab^3$$

$$\Rightarrow (a+b)^4 - (a-b)^4 = 8ab(a^2 + b^2).$$

$$\text{Hence } (\sqrt{3} + \sqrt{2})^4 - (\sqrt{3} - \sqrt{2})^4 = 8\sqrt{3}\sqrt{2}(\sqrt{3}^2 + \sqrt{2}^2)$$

$$\Rightarrow (\sqrt{3} + \sqrt{2})^4 - (\sqrt{3} - \sqrt{2})^4 = 40\sqrt{6}.$$

Hence Result.

EXAMPLE



Find $(x + 1)^6 + (x - 1)^6$. Hence or otherwise evaluate $(\sqrt{2} + 1)^6 + (\sqrt{2} - 1)^6$.

h/w

EXAMPLE



Show that $9^{n+1} - 8n - 9$ is divisible by 64, whenever n is a positive integer.

H/W

EXAMPLE



Prove that $\sum_{r=0}^n 3^r nC_r = 4^n.$

Proof: From Binomial Theorem, we have;
 $(a+b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r. \text{---(1)}$

Setting $a=1$ & $b=3$ in (1), we get;

$$(1+3)^n = \sum_{r=0}^n {}^nC_r (1)^{n-r} (3)^r.$$

$$\Rightarrow \sum_{r=0}^n 3^r \cdot {}^nC_r = 4^n.$$

Hence proved.

QUESTION



If a and b are distinct integers, prove that $a - b$ is a factor of $a^n - b^n$, whenever n is a positive integer. [Hint: write $a^n = (a - b + b)^n$ and expand]

Proof: Here $a^n = (a - b + b)^n = (t + b)^n$, where $t = a - b$.

Using Binomial theorem, we get

$$a^n = {}^n C_0 t^n + {}^n C_1 t^{n-1} b + {}^n C_2 t^{n-2} b^2 + \dots + {}^n C_{n-1} t b^{n-1} + {}^n C_n b^n$$

$$\Rightarrow a^n = (a - b)^n + {}^n C_1 (a - b)^{n-1} b + {}^n C_2 (a - b)^{n-2} b^2 + \dots + {}^n C_{n-1} (a - b) b + b^n$$

$$\Rightarrow a^n - b^n = (a - b)^n + {}^n C_1 (a - b)^{n-1} b + \dots + {}^n C_{n-1} (a - b) b$$

$$\Rightarrow a^n - b^n = (a - b) \left\{ (a - b)^{n-1} + {}^n C_1 (a - b)^{n-2} b + \dots + {}^n C_{n-1} b \right\}$$

$$\Rightarrow a^n - b^n = (a - b) q$$

This shows that $a - b$ is a factor of $a^n - b^n$, where n is a positive integer.

Hence proved.

Thank
You